

BRAC UNIVERSITY

Department of Computer Science and Engineering

Examination: Quiz - 1
Duration: 25 minutes

Semester: Spring 2026
Full Marks: 20

CSE 340: Computer Architecture

Name: <i>Solution</i>	ID:	Section:
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1.

Instruction Set:

Add

Sub

Ld

Add

Sd

Sub

Sub

Sll

Add

Sub

Sub

Sub

Ld

Sd

Sub

Sll

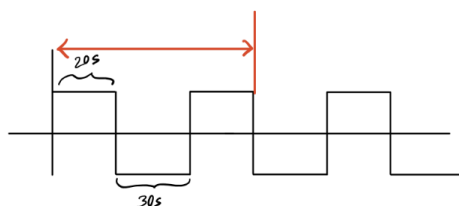


Figure: Sll Instruction

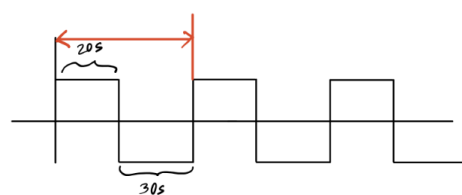


Figure: Ld Instruction

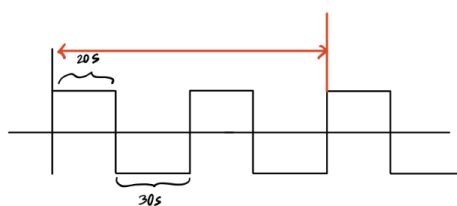


Figure: Sd Instruction

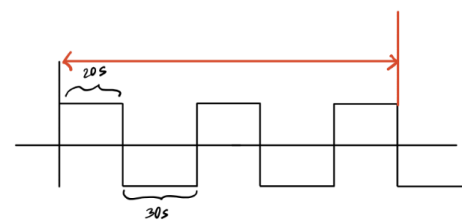


Figure: Sub Instruction

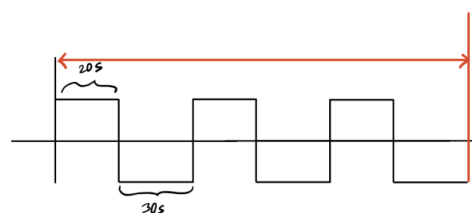


Figure: Add Instruction

a. Find CPIs; You must answer in the following table. [1 x 5]

	Add	Sub	Ld	Sd	Sll
CPI	3	2.5	1	2	1.5
IC	3	7	2	2	2

b. Find the Frequency in GHz. Answer: [1]

c. Find the time to run the instruction set. [4]

c)

$$\text{Avg. CPI} = \frac{(3 \times 3) + (2.5 \times 7) + (1 \times 2) + (2 \times 2) + (1.5 \times 2)}{16} = \frac{32.5}{16} = 2.03$$

$$\text{CPU Time} = 16 \times 2.03 \times 50 = 17250$$

b)

$$\begin{aligned} \text{Clock Cycle Time} &= (30 + 20) \text{ s} \\ &= 50 \text{ s} \\ \text{Freq.} &= \frac{1}{\text{Clock Cycle Time}} \\ &= \frac{1}{50} \text{ Hz} \\ &= 0.02 \times 10^{-9} \text{ GHz} \end{aligned}$$

2. A personal computer is designed so that opening the programs you use frequently like the web browser or file explorer runs very quickly. Less-used programs, such as running system checks or disk cleanup, take a bit longer. **By making the frequently used programs faster, the PC feels faster and more responsive overall.**

Write the name of the relevant idea from the 7 Great Idea(s) that is applied in the given scenario. [2]

Answer: **Make the common case faster**

3. Computer A has a MIPS of 1500. You run a program on this computer which has an Instruction count of 15×10^9 .

a) Find the CPU execution time of the program. [2]

b) Further analysis shows that 60% of the total execution time is spent executing Sub instructions and 30% on Add instructions. You want to improve the performance of Computer A so that the overall execution time becomes 5s by only improving the performance of the Sub instruction. T_{new}

Determine how much the Sub instructions must be sped up to achieve the target. [6]

$$a) \text{ MIPS} = \frac{IC}{\text{Exe. Time} \times 10^6}$$

$$\Rightarrow 1500 = \frac{15 \times 10^9}{\text{Exe. Time} \times 10^6}$$

$$\Rightarrow 1500 = \frac{15 \times 10^3}{\text{Exe. Time}}$$

$$\Rightarrow \text{Exe. Time} = 10s$$

$$b) T_{new} = \frac{T_{aff}}{n} + T_{unaff}$$

$$\Rightarrow 5 = \frac{6}{n} + 4$$

$$\Rightarrow \frac{6}{n} = 1$$

$$\Rightarrow n = 6$$

$$MIPS = 1500$$

$$IC = 15 \times 10^9$$

$$T_{new} = 5s$$

$$T_{aff} = (10 \times 0.6)s$$

$$= 6s$$

$$T_{unaff} = (10 - 6)s$$

$$= 4s$$